

Some “sophisticated” Category

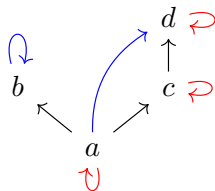
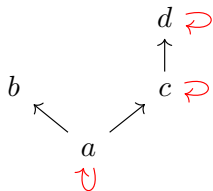
Learning outcomes

- Free categories from graphs
- Categorical monoids

Free categories from graphs

An example and the generalization

Example: from a graph to a category



Definition of the free category $\mathcal{F}(G)$ from/on G .

For every $G = (V, E)$:

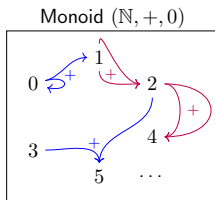
- $\text{obj}(\mathcal{F}(G)) = V$
- $\mathcal{F}(G)(v, u) = \left\{ v \xrightarrow{f} u \mid v \neq u \wedge \{(v, v_1), \dots, (v_n, u)\} \subseteq E \right\}$
- $\mathcal{F}(G)(v, v) = \left\{ v \xrightarrow{\text{id}_v} v \mid v \in V \right\}$

Categorical monoids

The driving experiment to change perspective

- `Support.CT0100FreeMonoidSum_ToFill.hs`
- `Support.CT0100FreeMonoidSum.hs`
(restricted)

The change of perspective, but formally



The categorical monoid $\mathcal{N}$ from $(\mathbb{N}, +, 0)$.

- $\text{obj}(\mathcal{N}) = \{\bullet\}$, $\text{hom}(\mathcal{N}) = \{(0+), (1+), (2+), \dots \mid (n+) : \bullet \rightarrow \bullet\}$
- axiom to compose section-elements into a section-element:

$$\bullet \xrightarrow{(m+)} \bullet \xrightarrow{(n+)} \bullet \quad \text{with a curved arrow from } \bullet \text{ to } \bullet \text{ labeled } (n+m+)$$

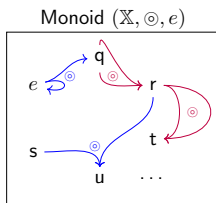
- $(0+)$ is the identity:

$$\bullet \xrightarrow{(0+)} \bullet \xrightarrow{(n+)} \bullet \quad \text{with a curved arrow from } \bullet \text{ to } \bullet \text{ labeled } (n+)$$

$$\bullet \xrightarrow{(m+)} \bullet \xrightarrow{(0+)} \bullet \quad \text{with a curved arrow from } \bullet \text{ to } \bullet \text{ labeled } (m+)$$

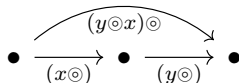
- $+$ associative $\Rightarrow$ composition associative in $\mathcal{N}$

The generalization on a monoid $(X, \odot, e)$

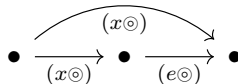
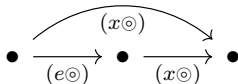


Categorical monoid $\mathcal{X}$

- $\text{obj}(\mathcal{X}) = \{\bullet\}$, $\text{hom}(\mathcal{X}) = \{(e \odot), (q \odot), (r \odot), \dots \mid (x \odot) : \bullet \rightarrow \bullet\}$
- axiom to compose section-elements into a section-element:



- $(e \odot)$ is the identity:



- $\odot$ associative $\Rightarrow$ composition associative in $\mathcal{X}$

Write the Categorical monoid of the following monoids:

- $(\mathbb{B}, \wedge, \mathbf{T})$
Support.CT0100FreeMonoidBooleanAndCat
- $(\mathbb{B}, \vee, \mathbf{F})$
- $(\mathbb{B}, \oplus, \mathbf{F})$ (“exclusive or”)
- $(\Sigma^*, \cdot, \epsilon)$ (strings)
- $(\{[a_1, \dots, a_n] \mid n \in \mathbb{N}\}, ++, [\])$ (lists)

where $\mathbb{B} = \{\mathbf{T}, \mathbf{F}\}$



Milewski, B. and I. Tabachnik (2019). *Category Theory for Programmers*.

<https://bartoszmilewski.com/2014/10/28/category-theory-for-programmers-the-preface/>. Lightning Source UK. ISBN: 9780464243878. URL:

<https://books.google.it/books?id=hA55yQEACAAJ>.